BIPHMUN 2026

Background Guide

Committee: General Assembly

Topic: Mitigating Active Debris and Regulating Future Launches to Ensure Space Sustainability

Agenda Overseen by: Crystal Wang

Letter from the Chair

Dear BIPHMUN 2026 General Assembly delegates,

As President Chair of the General Assembly, I am thrilled to welcome each and every one of you to BIPHMUN 2026, and I am excited to be working with you over the course of this conference to ensure a smooth, fruitful experience. The General Assembly will be debating two topics over three days: regulating the use of cryptocurrency in illegal drug trafficking activities; mitigating active debris and regulating future launches to ensure space sustainability. Delegates are expected to prepare an opening speech for each topic detailing your country's stance and interests, conduct thorough research, and begin to consider the formation of blocs for the process of resolution writing. I hope that you will all have a chance to grow, no matter your previous experience in MUN, and take away positive memories by the conclusion of this conference.

Happy debating!

Introduction

The General Assembly of the UN serves as the main policy-making committee, comprised of all 193 member states. The GA focuses on issues of international law and human rights, providing an inclusive platform for international decision-making and cooperation. It is further in charge of approving the UN budget, appointing the Secretary-General, and electing the non-permanent members of the Security Council. Over the course of its 80-year history, the General Assembly has enacted multiple major resolutions and actions, including the creation of UNICEF (1946), the adoption of the Universal Declaration of Human Rights (1948), the establishment of the 2030 Agenda for Sustainable Development, and more. Recently, GA has mainly focused on addressing the conflicts occurring in Ukraine and Gaza, as well as establishing sustainable guidelines for the use and development of artificial intelligence (AI).

The issue of space sustainability and active debris presents an increasing problem as technology and modern developments facilitate the widespread involvement of nation-states in space exploration. While significant progress has been made in this development, it has also simultaneously given rise to the accumulation of space debris. NASA defines space junk, or space debris, as mostly "human-generated objects", including big objects such as pieces of spacecraft, nonfunctional satellites, and parts of a rocket, as well as smaller things such as flecks of paint from various manmade space objects. Space debris may also be generated from explosions of objects in orbit moving at high speeds.

Consequently, 60 years of space activity has resulted in more than 6050 launches, with about 28160 tracked objects remaining in space. Currently, there are close to 6,000 tons of material in low earth orbit (LEO), and yet no binding international space laws exist enacting regulations on space debris in LEO or establish methods to clean it up.

The launch of Sputnik I by the Soviet Union on October 4, 1957, marked the start of decades of satellite activity and the development of rockets, while also marking the start of the rapid accumulation of manmade objects in space. Throughout the 20th century, both robotic and human space exploration grew exponentially, officially launching the space age. On June 29, 1961, a Thor-Ablestar rocket upper stage was reported to have exploded, marking the very first satellite blowup and creating more than 200 identified fragments. After Sputnik, between 50 and 100 satellites were launched into space every year, a trend that continued until the 2010s. Since then, various private firms such as SpaceX have been established to drive innovation and build the required technology. In 2024, rocket launches averaged one every 34 hours, overall sending more than 2,800 satellites into orbit. As of early 2026, there are approximately 15,000 artificial satellites orbiting Earth, with the majority active and in LEO. The International Space Station (ISS) resides at an altitude of approximately 250 miles (400 kilometers) up, and at this height, orbital velocity is around 17,100 miles per hour. At this tremendous speed, even tiny fragments of debris can inflict serious damage to functional spacecrafts, posing a serious problem to the future development of space exploration.

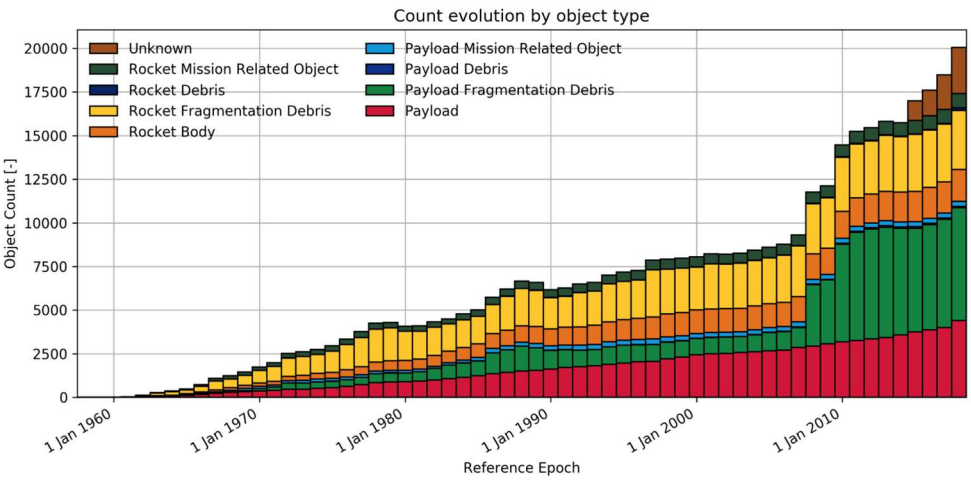

Figure 1. The European Space Agency

The buildup of active space debris has led to the coining of the term "Kessler Syndrome", inspired by the 1978 paper "Collision frequency of artificial satellites: The creation of a debris belt" written by Donald Kessler, a researcher in NASA's Environmental Effects Office. Although the formal definition of this term remains open to debate, it usually refers to a chain reaction phenomenon where the amount of debris in orbit around Earth becomes dense to the point where collisions between these objects cascade, exponentially increasing space pollution over time. Eventually, space junk would multiply to a point where functional satellites are destroyed and further exploration becomes impossible, hindering human space ambitions in the long run.

The 1967 Outer Space Treaty (OST) established several guidelines that provide the foundation for current international space law, including restricting the use of outer space and celestial bodies solely for peaceful purposes, mandating that states be responsible for all damage caused by their space objects, and ordering the avoidance of harmful

contamination. However, while the OST forbids the national ownership of celestial bodies, this opens the opportunity for private resource extraction and exploitation. Furthermore, the OST is outdated and in urgent need of amendment to maintain relevance to the current global situation regarding space exploration.

While the accumulation of space debris entails serious consequences, overregulation may lead to blocks on innovation, slowing the progression of spacefaring projects. The rise of privatized firms seeking to create profit from engaging in space activity adds another layer of complexity to developing regulations for future launches. SpaceX launched over 3,000 satellites in the year 2025, contributing to about 85% of all American orbital launches in the entire year. With regulations placed on satellite activity in order to lessen the burden of space debris, progression in space exploration as well as the growth of private companies such as SpaceX could be severely limited. Strict spacefaring rules may also critically limit the advancement of space exploration efforts in developing countries, inadvertently raising barriers to entry by increasing costs and technological restrictions. Compliance to regulations such as the IADC Space Debris Mitigation Guidelines, while representing internationally recognized practices to mitigate space debris, can impose a significant financial burden on developing countries. For example, defunct satellites and spacecraft in LEO were dictated to be disposed of within 25 years, known as the "25-year rule". This naturally calls for more sophisticated and advanced technology to be developed before spacefaring actions can be taken in order to ensure adherence to space debris disposal rules. Such regulations may also lead to a "first come, first serve" issue, where policy favors countries currently leading the space industry. By the time smaller countries are able to join the space race, optimal orbits such as LEO may have already been deemed unsustainable for further launches.

Finally, another complication arises from the question of who or what entity the responsibility of the removal of existing space debris should fall upon. Article VII of the Outer Space Treaty states that countries will be held internationally liable for the damage caused by space objects they have launched. The subsequent Liability Convention also mandated countries responsible for damage caused by their space objects on the surface of Earth. However, minimal to no legal consequences remain for noncompliance, and no guidelines exist that work to resolve issues arising from space debris damaging another party's spacecraft or satellite. Definitions for what qualifies as damage are vague, and it is difficult to prove responsibility.

Definition of Key Terms

Space Debris

All non-functional, human-made objects, including fragments and elements thereof, in Earth orbit or re-entering into Earth's atmosphere.

Low Earth Orbit (LEO)

An orbit around Earth with an altitude that lies towards the lower end of the range of possible orbits. This is around 1,200 miles (2,000 kilometers) or less.

Kessler Syndrome

A situation in which the density of objects in low Earth orbit becomes so high due to space pollution that collisions between these objects cascade, exponentially increasing the amount of space debris over time.

Active Debris Removal (ADR)

The deliberate, on-orbit intervention to capture, remove, or de-orbit non-functional human-made objects from Earth's orbit, particularly from crowded regions like LEO.

Passivation

The process of removing stored energy from a space vehicle to reduce the risk of high-energy releases (e.g., explosions, fragmentations, etc.) that could produce orbital debris after the end of a mission.

Megaconstellation

A large, coordinated network of satellites operating together for the purpose of delivering broadband Internet access.

Committee on the Peaceful Uses of Outer Space (UN COPUOS)

A United Nations committee set up by the General Assembly to govern the exploration and use of space for the benefit of all humanity, for peace, security, and development.

Past International Actions

1979	The NASA Orbital Debris Program officially began in 1979, representing one of the earliest national efforts at mitigating the issue of space debris. The program looked to design equipment and new technology that can track and remove the debris already present in space. While largely considered successful as it set important initiatives and mitigation guidelines, it ultimately failed to stop the net increase of space debris in the decades to come, suffering from a shortage in funds as well as an inability to keep up with heavy responsibilities corresponding to the rapid increase in orbital space objects.
1993	The Inter-Agency Space Debris Coordination Committee (IADC) was established as an initial international platform centered around the mitigation of space debris and the promotion of space sustainability. It included key players in the industry such as the USA, China, and Russia. However, the IADC suffers from the crucial weakness of noncompliance, as its guidelines are voluntary and non-binding.
1999	The UN Committee on the Peaceful Uses of Outer Space (UN COPUOS) releases its first *Technical Report on Space Debris,* being the first comprehensive report to fully detail the outer space debris environment, while providing a risk assessment that highlights the threat to space sustainability. Its main shortcoming again rests on the fact that it merely offers guidelines without establishing legally binding international law, thus limiting its effectiveness.
2007	The *Space Debris Mitigation Guidelines* developed by UN COPUOS is adopted by the General Assembly. While representing a crucial step towards the creation of an international consensus on the mitigation of space debris, it fails to fully regulate space activity with mandatory rules.
2012	The European Space Agency (ESA) launches the Clean Space initiative that aims to assist technological development in debris removal, passivation, and ensuring that satellites burn up upon reentry into Earth's atmosphere. Nonetheless, it suffers from financial, technical, and geopolitical challenges as a result of attempting to resolve an issue of such immense scale.
2023-	ESA develops its "Zero Debris" approach, outlining guidelines that aim to

2024	stop the creation of new space debris by 2030.

In recent years, organizations and policymakers have gradually shifted regulatory focus to introduce stricter, mandatory guidelines in an attempt to increase the effectiveness of policies. Efforts in the past, such as the IADC and UN COPUOS guidelines, were mainly limited in their ability to enforce compliance. Additionally, clear, technical definitions of ambiguous terms or phrasing are often lacking, introducing legal complications that hinder the progression of space debris mitigation efforts.

Active Debris Removal (ADR), technologies or expeditions designed to physically capture or remove existing debris from orbit, have been steadily on the rise. The ESA's ClearSpace-1 mission, set to launch in 2028, would become the world's first ever active debris removal mission, targeting the removal of the PROBA-1 satellite from orbit. Japan's JAXA has launched the Commercial Removal of Debris Demonstration (CRD2) program with the purpose of implementing ADR as a new private business market, thus increasing funds and allowing the private sector to demonstrate ADR technology and actively participate in mitigation efforts.

Bloc Stances

Established Space Powers

Market-driven

Countries such as the United States, United Kingdom, Australia, and Canada have traditionally advocated for market-driven innovation, flexible frameworks, and non-binding guidelines that encourage the growth of the private sector. Private companies such the United States' SpaceX and Blue Origin work to facilitate technological advancement, serving as the key drivers of space exploration. The Artemis Accords, implemented in October 2020 by NASA and the U.S. Department of State serves as an exemplary non-binding framework encouraged by these countries that work to guide sustainable and peaceful space activity.

State-led

State-focused established space powers include countries such as China and Russia that uphold multilateral frameworks and encourage the development of legal, mandatory policies negotiated within the UN COPUOS. Sovereign control and responsibility over all spacefaring activities is prioritized and emphasized, and the potential weaponization of debris-removal technologies is of major concern, leading to strict demands for international oversight in order to prevent the escalation of military conflicts in space.

Middle Powers

Possessing adequate levels of technological and financial abilities, middle power countries such as Japan, South Korea, and the United Arab Emirates primarily advocate for intermediate solutions, favoring guidelines with enough binding power to enforce compliance while leaving flexibility to allow for the evolution of space technology. They mainly support the establishment of international standards and peaceful cooperation, facilitating the coexistence of responsible and innovative behavior in the space industry.

Rising Space Powers

Comprised of countries such as France, Germany, and India, the developing space powers further support intermediate policies, emphasizing the ethical and safe development of

technology, transparency of data, and multilateral cooperation through platforms such as the ESA. The Indian Space Policy of 2023 outlined the responsibilities of non-governmental entities, aiming to promote human presence in space in collaboration with both domestic and international stakeholders.

Safeguards of Equity and Space Accessibility

Countries including Brazil, Nigeria, and Singapore aim primarily to preserve the equal accessibility of space to all nations regardless of economic and technological development levels. They seek to warn against the proliferation of mega-constellations such as Starlink that threaten to hinder the growth of emerging space programs, especially in developing countries. Brazil has traditionally supported the Common Heritage of Mankind principle, which states that global areas and resources like outer space are held for the benefit of humanity as a whole instead of being open to claims of sovereignty. These nations further support a space debris mitigation fund to be financed by historically debris-generating nations.

Potential Solutions

The mitigation of space debris and the development of regulatory frameworks present a complex, dynamic issue that calls for a multifaceted approach dependent on the unified effort of member states. To begin with, technological innovation serves as the first and foremost driver in the mitigation of space debris. Technology may be developed for the purpose of the removal of existing space debris, including robotic arms, nets, harpoons, and magnetic mechanisms that are able to secure space objects and ensure a controlled reentry into the atmosphere. Contactless removal methods are further in the testing, including techniques such as laser ablation, which uses laser systems to vaporize small portions of debris, generating a high-velocity plasma plume that serves to alter the object's orbit. Another category of technology focuses on proactive prevention measures that aim to minimize the amount of debris a satellite launched into space would generate. The "Zero Debris" approach proposed by the ESA works to utilize materials that can vaporize completely in the atmosphere during re-entry, preventing debris from reaching the ground. Future techniques similar to passivation can serve to mitigate accidental explosions of spacecraft and satellites, greatly reducing the debris a space object can generate. Continued development of technology that can detect incoming debris, maneuvering the satellite out of the way in time, may prove to be effective in preventing debris caused by collisions.

As previously addressed, the major shortcoming of past international efforts lies in the fact that the majority of existing guidelines are "soft laws" instead of being mandatory or legally binding. Gradually progressing towards stricter frameworks of regulation may improve the success of existing policies which establish significant rules but fail to enforce compliance. The development of a clear legal framework detailing compensation and punishment measures for damage caused by space debris may also serve to encourage innovation from governments and private firms for more sophisticated technology.

Questions to Answer

Should policy be developed to prioritize preventing debris or removing it afterward?

How should the responsibility of removing existing space debris be distributed?

What are the consequences of implementing legally binding space debris mitigation laws?

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